Cloud Computing

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Defining the Cloud

- Term cloud has been used historically as a metaphor for the Internet
- This usage was originally derived from its common depiction in network diagrams as an outline of a cloud
- Cloud itself is a set of hardware, networks, storage, services, and interfaces that enable the delivery of computing as a service
- Cloud services include the delivery of software, infrastructure, and storage over the internet
- A Cloud is a type of parallel and distributed system consisting of a collection of interconnected and virtualized computers that are dynamically provisioned and presented as one or more unified computing resources based on service-level agreements established through negotiation between the service provider and consumers.

Defining Cloud Computing

- "The National Institute of Standards and Technology (NIST) defines cloud computing as a "pay-per-use model for enabling available, convenient and on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction."

- Cloud computing is using the internet to access someone else's software running on someone else's hardware in someone else's data center.

- A large-scale distributed computing paradigm that is driven by economies of scale, in which **a pool of abstracted, virtualized, dynamically scalable, managed computing power, storage, platforms, and services** are delivered on demand to external customers over the Internet.

What is Cloud Computing?

- **Cloud Computing** is a general term used to describe a new class of network based computing that takes place over the Internet,
  - basically a step on from Utility Computing
  - a collection/group of integrated and networked hardware, software and Internet infrastructure (called a platform).
  - Using the Internet for communication and transport provides hardware, software and networking services to clients
- These platforms hide the complexity and details of the underlying infrastructure from users and applications by providing very simple graphical interface or API (Applications Programming Interface).

What is Cloud Computing?

- In addition, the platform provides on demand services, that are always on, anywhere, anytime and any place.
- Pay for use and as needed, elastic
  - scale up and down in capacity and functionalities
- The hardware and software services are available to
  - general public, enterprises, corporations and businesses markets

Cloud Computing

- Cloud computing is the use of computing resources (hardware and software) that are delivered as a service over a network (typically the Internet).
- Cloud computing is a general term for anything that involves delivering hosted services over the Internet.
- These services are broadly divided into three categories: Infrastructure-as-a-Service (IaaS), Platform-as-a-Service (PaaS) and Software-as-a-Service (SaaS).
- Delivery of computational resources from a location other than the one from which you are computing.

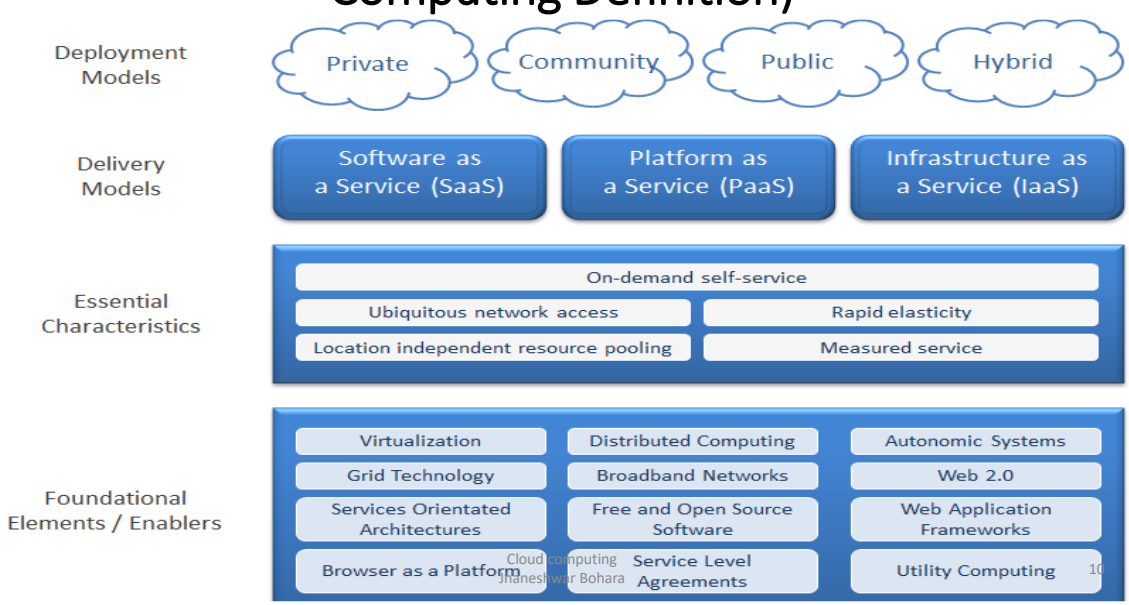
Emergence of Cloud Computing

- The underlying concept of cloud computing dates back to the 1950s; when large-scale mainframe became available in academia and corporations, accessible via thin clients /terminal computers
- The cloud symbol was used to represent the Internet as early as 1994.
- In 1997, Professor Ramnath Chellapa of Emory University and the University of South California defined cloud computing as the new "computing paradigm where the boundaries of computing will be determined by economic rationale rather than technical limits alone."

Emergence of Cloud Computing

- **One of the first milestones for cloud computing was the arrival of Salesforce.com in 1999**, which pioneered the concept of delivering enterprise applications via a simple website.
- **The next development was Amazon Web Services in 2002,** which provided a suite of cloud-based services including storage, computation and even human intelligence through the Amazon Mechanical Turk.
- Then in 2006, Amazon launched its Elastic Compute cloud (EC2) as a commercial web service that allows small companies and individuals to rent computers on which to run their own computer applications.

Cloud at a Glance (NIST Visual Model of Cloud Computing Definition)



Cloud Computing Characteristics

Common Characteristics:

- Massive Scale
- Resilient Computing
- Homogeneity
- Geographic Distribution
- Virtualization
- Service Orientation
- Low Cost Software
- Advanced Security

Essential Characteristics:

- On Demand Self-Service
- Broad Network Access
- Rapid Elasticity
- Resource Pooling
- Measured Service

Essential Characteristics

- On-demand self-service
  - A consumer can unilaterally provision computing capabilities such as server time and network storage as needed automatically, without requiring human interaction with a service provider(Self-service provisioning and automatic de provisioning)
- Broad network access
  - Capabilities are available over the network and accessed through standard mechanisms that promote use by heterogeneous thin or thick client platforms (e.g., mobile phones, laptops, and PDAs) as well as other traditional or cloud based software services.

Essential Characteristics

- Resource pooling
  - The provider's computing resources are pooled to serve multiple consumers using a multi-tenant model, with different physical and virtual resources dynamically assigned and reassigned according to consumer demand.
- Rapid elasticity
  - Capabilities can be rapidly and elastically provisioned - in some cases automatically - to quickly scale out; and rapidly released to quickly scale in(Elasticity and the ability to scale up and down)
  - To the consumer, the capabilities available for provisioning often appear to be unlimited and can be purchased in any quantity at any time.

Essential Characteristics

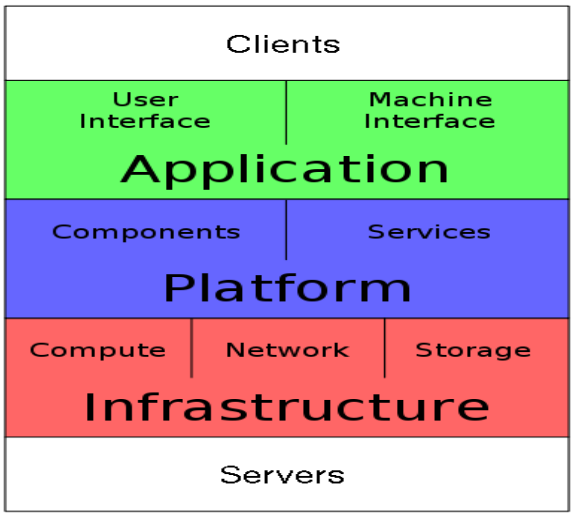
- Measured service
  - Cloud systems automatically control and optimize resource usage by leveraging a metering capability at some level of abstraction appropriate to the type of service.
  - Resource usage can be monitored, controlled, and reported - providing transparency for both the provider and consumer of the service.
  - Billing and metering of service usage in a pay-as-you-go model

Cloud Service Models

SPI Model

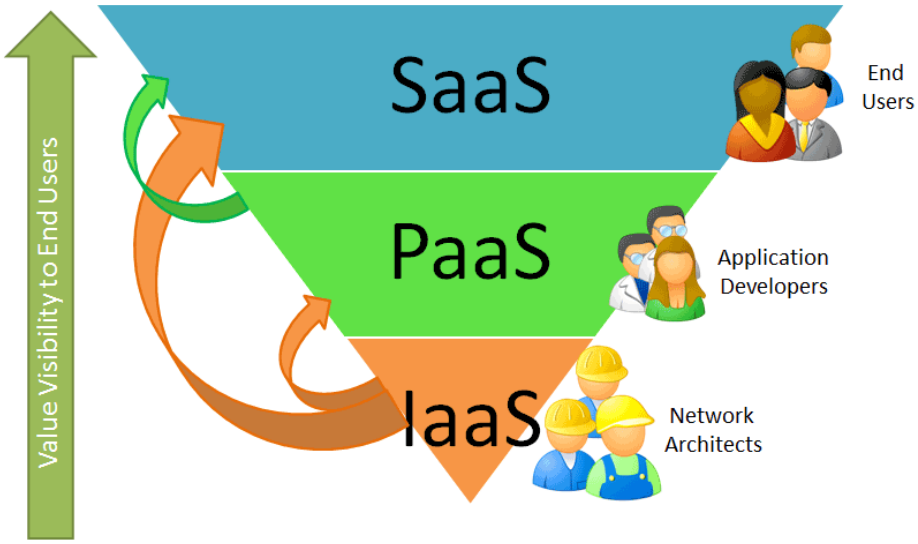
- Cloud **S**oftware as a Service (**S**aaS)
- Cloud **P**latform as a Service (**P**aaS)
- Cloud **I**nfrastructure as a Service (**I**aaS)

Cloud Service Models



Cloud Computing Stack

Cloud Computing Layers



Infrastructure as a Service (IaaS)

- The capability provided to the consumer is to provision processing, storage, networks, and other fundamental computing resources.
- Consumer is able to deploy and run arbitrary software, which can include operating systems and applications.
- The consumer does not manage or control the underlying cloud infrastructure but has control over operating systems, storage, deployed applications, and possibly limited control of select networking components (e.g., host firewalls).
- Examples: Amazon EC2, GoGrid, iland, Rackspace Cloud Servers, ReliaCloud.

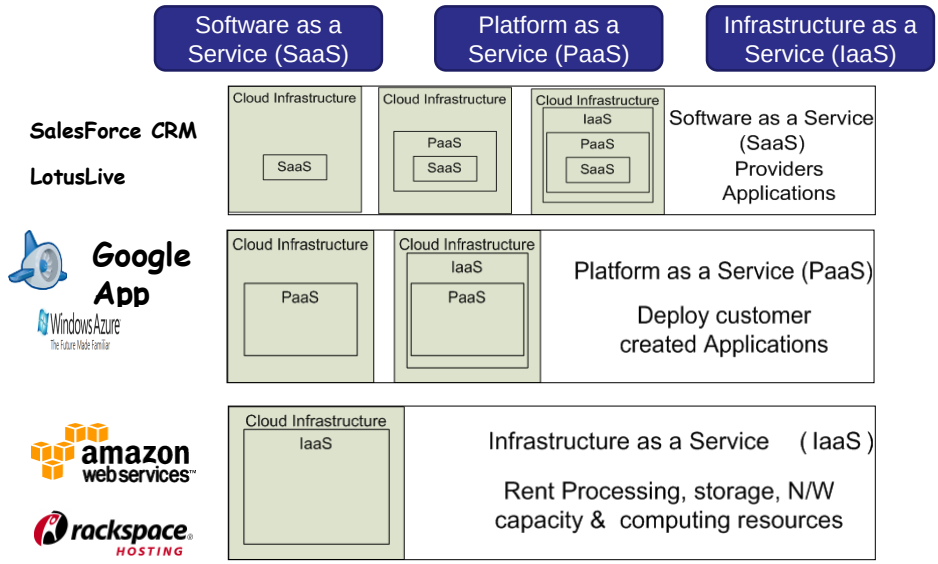
Platform as a Service (PaaS)

- The capability provided to **the consumer is to deploy onto the cloud infrastructure consumer created or acquired applications** created using programming languages and tools supported by the provider.
- The consumer does not manage or control the underlying cloud infrastructure including network, servers, operating systems, or storage, but has control over the deployed applications and possibly application hosting environment configurations.
- ***Examples: Windows Azure, Google App Engine.***

Software as a Service (SaaS)

- The capability provided to the consumer is to use the provider's applications running on a cloud infrastructure.
- The applications are accessible from various client devices through a thin client interface such as a web browser (e.g., web-based email).
- The consumer does not manage or control the underlying cloud infrastructure including network, servers, operating systems, storage, or even individual application capabilities, with the possible exception of limited user specific application configuration settings.
- ***Examples: Caspio, Google Apps, Salesforce, Nivio, Learn.com.***

Cloud Service Models



Cloud Deployment Models

- Public Cloud

  The cloud infrastructure is made available to the general public or a large industry group and is owned by an organization selling cloud services.

- Private Cloud

  The cloud infrastructure is operated solely for a single organization. It may be managed by the organization or a third party, and may exist on-premises or off-premises.

Cloud Deployment Models

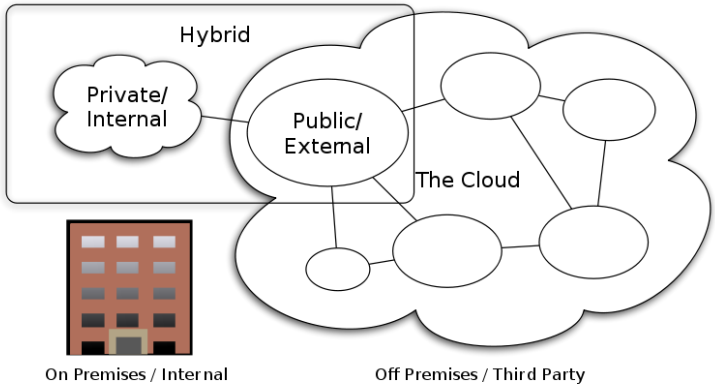
- Community Cloud

  The cloud infrastructure is shared by several organizations and supports a specific community that has shared concerns (e.g., mission, security requirements, policy, or compliance considerations). It may be managed by the organizations or a third party and may exist on-premises or off-premises.

- Hybrid Cloud

  The cloud infrastructure is a composition of two or more clouds (private, community, or public) that remain unique entities but are bound together by standardized or proprietary technology that enables data and application portability (e.g., cloud bursting for load- balancing between clouds).

Cloud Deployment Models



Cloud Computing Types

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Components of Cloud Computing

- The Client- The End User
- The Service (Functions in Cloud Computing)
- The Application
- The Platform
- The Storage

Benefits of Cloud Computing

- Business Benefits of Cloud Computing
  - Almost zero upfront infrastructure investment
  - Just-in-time Infrastructure
  - More efficient resource utilization
  - Usage-based costing
  - Reduced time to market
- Technical Benefits of Cloud Computing
  - Automation – “Scriptable infrastructure”
  - Auto-scaling
  - Proactive Scaling
  - More Efficient Development lifecycle
  - Improved Testability
  - Disaster Recovery and Business Continuity

Legal Issues in Cloud Computing

- Confidentiality
- Liability and responsibility
- Compliance
- Data protection, safety and recovery
- Copyright and Ownership
- Data portability
- Right to Audit
- Right to Audit

Ethical issues in cloud computing

- Cloud computing is based on a paradigm shift with profound implications on computing ethics.
- The main elements of this shift are:
  - the control is relinquished to third party services
  - the data is stored on multiple sites administered by several organizations
  - multiple services interoperate across the network
- Unauthorized access, data corruption, infrastructure failure, or unavailability are some of the risks related to relinquishing the control to third party services
- Moreover, it is difficult to identify the source of the problem and the entity causing it

Challenges for cloud computing

- **Security and Privacy**
  - Perhaps two of the more "hot button" issues surrounding cloud computing relate to storing and securing data, and monitoring the use of the cloud by the service providers.
  - These challenges can be addressed, for example, by storing the information internal to the organization, but allowing it to be used in the cloud
  - For this to occur, though, the security mechanisms between organization and the cloud need to be robust and a Hybrid cloud could support such a deployment

Challenges for cloud computing

- **Lack of Standards**
  - Clouds have documented interfaces; however, no standards are associated with these, and thus it is unlikely that most clouds will be interoperable
  - The Open Grid Forum is developing an Open Cloud Computing Interface to resolve this issue and the Open Cloud Consortium is working on cloud computing standards and practices

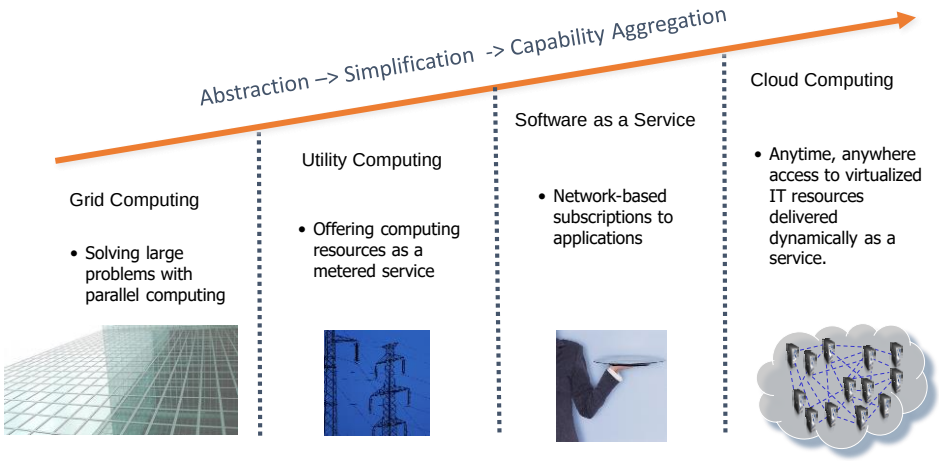
Challenges for cloud computing

- **Continuously Evolving**
  - User requirements are continuously evolving, as are the requirements for interfaces, networking, and storage
  - This means that a "cloud," especially a public one, does not remain static and is also continuously evolving.

Challenges for cloud computing

- **Compliance Concerns**
  - The Sarbanes-Oxley Act (SOX) in the US and Data Protection directives in the EU are just two among many compliance issues affecting cloud computing, based on the type of data and application for which the cloud is being used.
  - The EU has a legislative backing for data protection across all member states, but in the US data protection is different and can vary from state to state.
  - As with security and privacy mentioned previously, these typically result in Hybrid cloud deployment with one cloud storing the data internal to the organization.

Evolution of Cloud Computing



Grid Computing

- Grid computing can be defined to be a methodology to provide the users a virtual supper computer by forming an infrastructure of software, and hardware through clustering technique utilizing databases, networks, scientific instruments, computing methodologies and software packages from many different resources, and locations where the user can work collaboratively.
- Grid computing is a computer network in which each computer's resources are shared with every other computer in the system
- Processing power, memory and data storage are all community resources that authorized users can tap into and leverage for specific tasks.

Grid Computing

- A grid computing system can be as simple as a collection of similar computers running on the same operating system or as complex as inter-networked systems comprised of every computer platform you can think of .
- A grid computer is connected through a super-fast network and share the devices like disk drives, mass storage, printers and RAM.
- In general, a grid computing system requires:
  - At least one computer, usually a server, which handles all the administrative duties for the system
  - A network of computers running special grid computing network software
  - A collection of computer software called middleware

Grid Computing or Cloud Computing?

- Grid computing is often confused with cloud computing.
- Grid computing is a form of distributed computing that implements a *virtual supercomputer* made up of a cluster of networked or Internetworked computers acting in unison to perform very large tasks.
- Many cloud computing deployments today are powered by grid computing implementations and are billed like utilities, but cloud computing can and should be seen as an evolved next step away from the grid utility model.

Cloud: just a new name for Grid?

- YES: the vision is the same
  - To reduce the cost of computing
  - To increase reliability
  - To increase flexibility by transitioning from self operation to third party

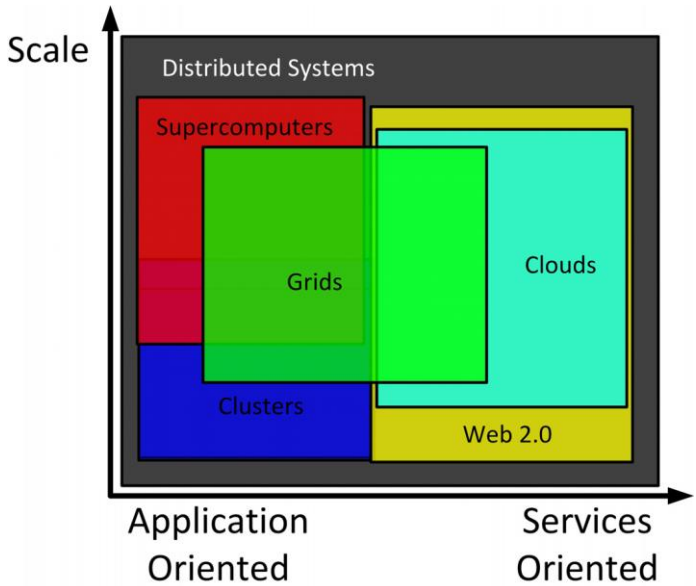
Cloud: just a new name for Grid?

- NO: things are different than they were 10 years ago
  - New needs to analyze massive data, increased demand for computing
  - Commodity clusters are expensive to operate
  - We have low-cost virtualization
  - Billions of dollars being spent by Amazon, Google, and Microsoft to create real commercial large-scale systems with hundreds of thousands of computers
  - The prospect of needing only a credit card to get on-demand access to infinite(100,000+) computers is exciting

Cloud: just a new name for Grid?

- Nevertheless YES: the problems are mostly the same
  - Common need to manage large facilities
  - Define methods to discover, request and use resources
  - Implement and execute highly parallel computations
  - Details differ, but issues are similar

Clouds, Grids and Distributed Systems



Clouds, Grids and Distributed Systems

- Web 2.0 covers almost the whole spectrum of service-oriented applications, where Cloud Computing lies at the large-scale side.
- Supercomputing and Cluster Computing have been more focused on traditional non-service applications.
- Grid Computing overlaps with all these fields where it is generally considered of lesser scale than supercomputers and Clouds.

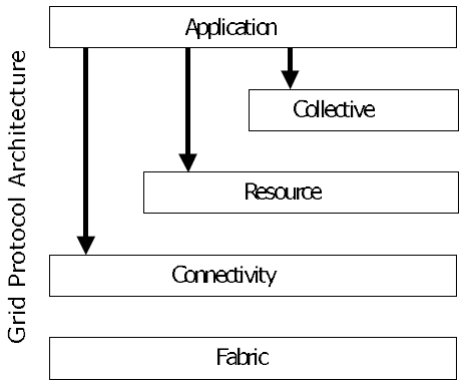
Comparing Grids and Clouds Side-by-Side

- Business model
- Architecture
- Resource Management
- Programming model
- Application model
- Security model

Comparing Grids and Clouds Side-by-Side: Business Model

- Grids:
  - Largest Grids funded by government
  - Largest user-base in academia and government labs to drive scientific computing
  - Project-oriented: service units
- Clouds:
  - Industry (i.e. Amazon) funded the initial Clouds
  - Large user base in common people, small businesses, large businesses, and a bit of open science research
  - Utility computing: real money

Comparing Grids and Clouds Side-by-Side: Architecture



Grid Protocol Architecture

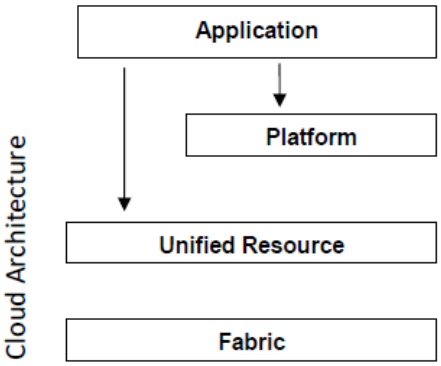


Figure 3: Cloud Architecture

Comparing Grids and Clouds Side-by-Side: Architecture(Grids)

- Grids provide protocols and services at five different layers as identified in the Grid protocol architecture
- At the ***fabric layer***, Grids provide access to different resource types such as compute, storage and network resource, code repository, etc.
- Grids usually rely on existing fabric components, for instance, local resource managers (i.e. PBS, Condor etc), General-purpose components such as GARA (general architecture for advanced reservation), and specialized resource management services such as Falkon.
- The ***connectivity layer*** defines core communication and authentication protocols for easy and secure network transactions.
- The GSI (Grid Security Infrastructure) protocol underlies every Grid transaction.

Comparing Grids and Clouds Side-by-Side: Architecture(Grids)

- The ***resource layer*** defines protocols for the publication, discovery, negotiation, monitoring, accounting and payment of sharing operations on individual resources.
- The GRAM (Grid Resource Access and Management)protocol is used for allocation of computational resources and for monitoring and control of computation on those resources, and GridFTP for data access and high-speed data transfer.
- The ***collective layer*** captures interactions across collections of resources, directory services such as MDS (Monitoring and Discovery Service) allows for the monitoring and discovery of VO resources, Condor-G and Nimrod-G are examples of co-allocating, scheduling and brokering services, and MPICH for Grid enabled programming systems, and CAS (community authorization service) for global resource policies.

Comparing Grids and Clouds Side-by-Side: Architecture(Grids)

- The ***application layer*** comprises whatever user applications built on top of the above protocols and APIs and operate in VO environments.
- Examples are Grid workflow systems and Grid portals(TeraGrid)

Comparing Grids and Clouds Side-by-Side: Architecture(Cloud)

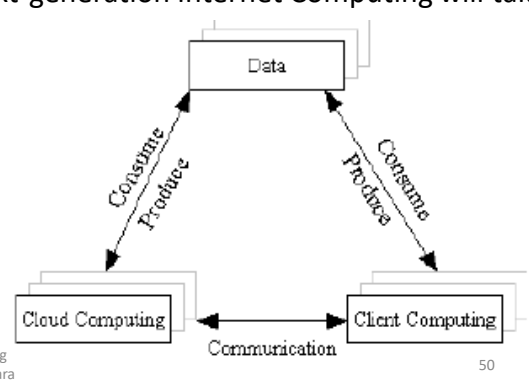
- The *fabric layer* contains the raw hardware level resources, such as compute resources, storage resources, and network resources.
- The *unified resource layer* contains resources that have been abstracted/encapsulated (usually by virtualization) so that they can be exposed to upper layer and end users as integrated resources, for instance, a virtual computer/cluster, a logical file system, a database system, etc.
- The *platform layer* adds on a collection of specialized tools, middleware and services on top of the unified resources to provide a development and/or deployment platform. For instance, a Web hosting environment, a scheduling service, etc.
- Finally, the *application layer* contains the applications that would run in the Clouds.

Comparing Grids and Clouds Side-by-Side: Resource management

- Compute model
  - Grids: batch-scheduled (queueing systems)
  - Clouds: resources shared by all users at the same time in contrast to dedicated resources in queueing systems
  - Maybe one of the major challenges in clouds: QoS

Comparing Grids and Clouds Side-by-Side: Resource management

- Data model:
  - People boldly predicate that future Internet Computing will be towards Cloud Computing centralized, in which storage, computing, and all kind of other resources will mainly be provisioned by the Cloud.
  - Foster, Zhao, Raicu and Lu envision that the next-generation Internet Computing will take the triangle model shown:
  - Internet Computing will be centralized around Data, Cloud Computing, as well as Client Computing.



Comparing Grids and Clouds Side-by-Side: Resource management

- Data model:
  - Grids: concept of virtual data, replica, metadata catalog, abstract structural representation
  - Data locality: to achieve good scalability data must be distributed over many computers and computations must be steered towards the best place to execute in order to minimize the communication costs
  - Clouds: Google's MapReduce system runs on top of the Google File System, within which data is loaded, partitioned into chunks, and each chunk replicated.
  - Grids: Data storage usually rely on shared file systems (NFS, GPFS, PVFS, Lustre) where data locality cannot be easily applied

Comparing Grids and Clouds Side-by-Side: Resource management

- Combining compute and data model:
  - Important to schedule computational tasks close to their data
  - Grids have been making progress in combining compute and data management with data-aware schedulers
  - Clouds will face significant challenges in handling data-intensive applications without serious efforts invested in harnessing the data locality of application access patterns

Comparing Grids and Clouds Side-by-Side: Resource management

- Virtualization:
  - Reasons for Virtualization: Abstraction and encapsulation
  - Clouds: rely heavily on virtualization
  - Grids: do not rely on virtualization as much as clouds. One example of use in Grids: Nimbus (previous Virtual Workspace Service)
  - Cloud Virtualization:
    - Server and app consolidation (multiple apps can run on the same server, resources can be utilized more efficiently)
    - Configurability
    - App availabillity (recovery)
    - Improved responsiveness
    - Meet SLA requirements

Comparing Grids and Clouds Side-by-Side: Resource management

- Monitoring:
  - Clouds: hard to do fine-control because of virtualization (problem for users and admins). In the future maybe not a problem as clouds become self-maintained and self-healing (autonomic)
  - Grids: several tools for monitoring (e.g. Ganglia)

Comparing Grids and Clouds Side-by-Side: Resource management

- Provenance:
  - Provenance refers to the derivation history of a data product, including all the data sources, intermediate data products, and the procedures that were applied to produce the data product.
  - Grids: built into a workflow system to support discovery and reproducibility of scientific results (Chimera, Swift, Kepler, VIEW etc)
  - Clouds: still unexplored
  - Scalable provenance querying and secure access to provenance info are still open problems for both grids and clouds

Comparing Grids and Clouds Side-by-Side: Programming model

- Grids: heavy use of workflow tools to be able to manage large sets of tasks and data. Focus on management rather than on interprocess communication, others: MPICH-G2, WSRF, GridRPC…
- Clouds: most use the map-reduce programming model. Implementation: Hadoop that uses Pig as a declarative programming language
- Clouds: Microsoft uses Cosmos (distributed storage system) and Dryad processing framework. DryadLINQ and Scope: declarative programming models
- Others: scripting languages: JavaScript, PHP, Python etc)
- Google App Engine uses Python as scripting language and GQL to query the BigTable storage system
- Interoperability: main challenge!

Comparing Grids and Clouds Side-by-Side: Application model

- Grids: generally support many different kinds of applications, ranging from high performance computing (HPC) to high throughput computing (HTC).
- Clouds: because of the use of virtualization may have difficulties in successfully running HPC applications that need fast and low latency networks
- Both grids and clouds have the capability to run any kind of application

Comparing Grids and Clouds Side-by-Side: Security model

- Clouds: seem to have a relatively simpler and less secure model than in grids, but virtualization gives a level of security
- Grids impose a stricter security model
- Cloud infrastructure typically rely on Web forms (over SSL) to create and manage account information for end-users, and allows users to reset their passwords and receive new passwords via Emails in an unsafe and unencrypted communication.
- Note that new users could use Clouds relatively easily and almost instantly, with a credit card and/or email address.
- To contrast this, Grids are stricter about its security.

Comparing Grids and Clouds Side-by-Side: Conclusions

- Clouds and Grids share a lot commonality in their vision, architecture and technology, but they also differ in various aspects such as security, programming model, business model, compute model, data model, applications, and abstractions.

Distributed Computing in Grid and Cloud

- **Distributed computing** is a field of computer science that studies distributed systems
- A distributed system consists of multiple autonomous computers that communicate through a computer network. The computers interact with each other in order to achieve a common goal.
- Distributed computing was one of the first real instances of cloud computing. Long before Google or Amazon, there was SETI@Home. Proposed in 1995 and launched in 1999, this program uses the spare capacity of internet connected machines to search for extraterrestrial intelligence. This is sort of the cloud in reverse.
- Distributed computing, as one can imagine, is where the computing elements of a network are spread over a large geographical area. Both cloud and grid computing are prime examples of distributed computing architectures.

Classification of Distributed Computing Systems

- These can be classified into 4 groups: clusters, peer-to-peer networks, grids, and clouds.
- A computing cluster consists of interconnected stand-alone computers which work cooperatively as a single integrated computing resource. The network of compute nodes are connected by LAN/SAN and are typically homogeneous with distributed control running Unix/Linux. They are suited to HPC.

Computational and Data Grids

- Grids are heterogeneous clusters interconnected by high-speed networks. They have centralized control, are server-oriented with authenticated security. They are suited to distributed supercomputing. E.g. TeraGrid.
- Like an electric utility power grid, a computing grid offers an infrastructure that couples computers, software/middleware, people, and sensors together.
- The grid is constructed across LANs, WANs, or Internet backbones at a regional, national, or global scale.
- The computers used in a grid include servers, clusters, and supercomputers. PCs, laptops, and mobile devices can be used to access a grid system.

Clouds

- A Cloud is a pool of virtualized computer resources. A cloud can host a variety of different workloads, including batch-style backend jobs and interactive and user-facing applications.
- Workloads can be deployed and scaled out quickly through rapid provisioning of VMs. Virtualization of server resources has enabled cost effectiveness and allowed cloud systems to leverage low costs to benefit both users and providers.
- Cloud system should be able to monitor resource usage in real time to enable rebalancing of allocations when needed.
- Cloud computing applies a virtualized platform with elastic resources on demand by provisioning hardware, software, and data sets dynamically. Desktop computing is moved to a service-oriented platform using server clusters and huge databases at datacenters.

Advantage of Clouds over Traditional Distributed Systems

- Traditional distributed computing systems provided for on-premise computing and were owned and operated by autonomous administrative domains (e.g. a company).
- These traditional systems encountered performance bottlenecks, constant system maintenance, poor server (and other resource) utilization, and increasing costs associated with hardware/software upgrades.
- Cloud computing as an on-demand computing paradigm resolves or relieves many of these problems.